

# Danesh Morales Hashemi

✉ [d2morale@uwaterloo.ca](mailto:d2morale@uwaterloo.ca)    [daneshmorales.github.io](https://daneshmorales.github.io)

Updated on October 2, 2026

## Research Positions

---

- 2025 – Present      **Graduate Research Assistant**, Institute for Quantum Computing  
Advisor: [Graeme Smith](#)
- 2024 – 2025      **Undergraduate Research Assistant**, Institute for Quantum Computing  
Advisor: [Joseph Emerson](#)  
Topic: *Circuit Benchmarking for Efficient and Scalable Fidelity Estimation*
- Fall 2023      **Undergraduate Research Assistant**, University of Waterloo  
Advisor: [Roberto Guglielmi](#)  
Topic: *Controllability of Fokker-Planck Equation*

## Education

---

- 2025 – Present      **MMath in Applied Mathematics**, University of Waterloo  
Quantum Information Specialization
- 2020 – 2025      **BMath in Mathematical Physics**, University of Waterloo  
Co-op Program • Pure Mathematics Minor • Graduated with [Dean's Honours](#)

## Conferences and Talks

---

- March 2025      **“Demonstrating Circuit Benchmarking for Scalable and Efficient Fidelity Estimation”**  
**APS March Meeting 2025**, Session on Scalable Verification and Validation (QCVV), Anaheim Convention Center

## Teaching Experience

---

- Spring 2025      **Teaching Assistant**, AMATH 250 – Introduction to Differential Equations
- Fall 2023      **Instructional Support Assistant**, MATH 115 – Linear Algebra for Engineering
- Winter 2022      **Instructional Support Assistant**, CS 115 – Introduction to Computer Science

## Projects

---

|               |                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| December 2025 | <b>“Introduction to Cycle Benchmarking for Quantum Error Diagnostics”</b><br>QIC 820 Final Project, Institute for Quantum Computing                        |
| December 2025 | <b>“Predicting MLB Playoffs Using Quantum Machine Learning”</b><br>QIC 890 Final Project, Institute for Quantum Computing. <a href="#">🔗 Link</a>          |
| April 2025    | <b>“Demonstrating Circuit Benchmarking for Efficient and Scalable Fidelity Estimation”</b><br>Undergraduate Thesis, University of Waterloo                 |
| August 2024   | <b>“Introduction to Superoperator Representations and Characterization of Completely Positive Maps”</b><br>AMATH 900 Final Project, University of Waterloo |

## Awards and Scholarships

---

|                                                                             |             |
|-----------------------------------------------------------------------------|-------------|
| Institute for Quantum Computing Undergraduate Research Award 2024           | 16,000 CAD  |
| University of Waterloo President’s Scholarship 2020                         | 2,500 CAD   |
| Full Undergraduate Educational Excellence Scholarship 2019, SENACYT, Panama | 250,000 USD |

## Relevant Coursework

---

(with links to syllabus)

|                                |                               |                         |
|--------------------------------|-------------------------------|-------------------------|
| Quantum Physics                | Quantum Theory 1/2/3          | Representation Theory   |
| Quantum Information Processing | Quantum Machine Learning      | Groups and Ring Theory  |
| Quantum Devices                | Theory of Quantum Information | Galois and Field Theory |